DEPARTMENT OF INDUSTRIAL MANAGEMENT

FACULTY OF SCIENCE | UNIVERSITY OF KELANIYA

INTE 22283 – Mobile Application Development

Week 1: Introduction to Mobile Computing and Mobile Networks

Sammani Maddumahewage

Lecturer (Probationary)

BSc (Hons) in Information Technology



Course Learning Outcomes (CLOs)

- **CLO 1:** Explain fundamental concepts of mobile computing, mobile networks, and major mobile OS architectures
- **CLO 2:** Compare native, hybrid, and cross-platform development approaches and select an appropriate platform
- **CLO 3:** Design user interfaces and user-flow prototypes applying UI/UX principles and navigation patterns
- **CLO 4:** Develop, debug, and test mobile applications integrating web services, sensors, and databases
- **CLO 5:** Implement security and privacy mechanisms including secure storage, authentication, and permissions
- **CLO 6:** Integrate mobile applications with IoT devices and evaluate deployment/monetization strategies

Today's Learning Outcomes

- Define mobile computing and identify its core characteristics
- Trace the evolution of mobile networks from 1G to 5G
- Explain cellular network architecture and mobility management
- Describe handoff, roaming, and location management concepts
- Differentiate mobile computing, wireless computing, and ubiquitous computing
- Identify the building blocks of the mobile application ecosystem
- Discuss current trends and industry relevance of mobile app development
- Outline module structure, assessment plan, and practical lab expectations

Why This Topic Matters

- **Ubiquity:** Over 7.2 billion smartphone subscriptions exist worldwide — mobile is the primary computing platform for most of humanity
- **Economic scale:** The global mobile app economy generates hundreds of billions of dollars annually through app stores, advertising, and services
- **Foundation for the module:** Every later topic — OS architecture, UI design, APIs, security, IoT — builds on the networking and platform concepts introduced today
- **Career relevance:** Mobile development is one of the most in-demand software engineering specializations in the current job market

What Is Mobile Computing?

- **Definition:** Mobile computing is a technology paradigm that allows transmission of data, voice, and video via a computer or any other wireless-enabled device, without being connected to a fixed physical link
- **Key idea:** The computing device and the user are free to move while remaining connected to a network and to data
- **Distinguishing factor:** Unlike traditional desktop computing, mobile computing assumes the device's location, connectivity, and available resources can change at any time

Core Characteristics of Mobile Computing

- **Portability**
  - Devices are small, lightweight, and battery-powered for movement
- **Connectivity**
  - Continuous or intermittent access to wireless networks (cellular, Wi-Fi, Bluetooth)
- **Interactivity**
  - Real-time, two-way communication between user, device, and remote services
- **Individuality**
  - Devices are typically personalized to a single user's preferences, identity, and context
- **Context-awareness**
  - Applications can sense and respond to location, motion, time, and environment

Components of a Mobile Computing System

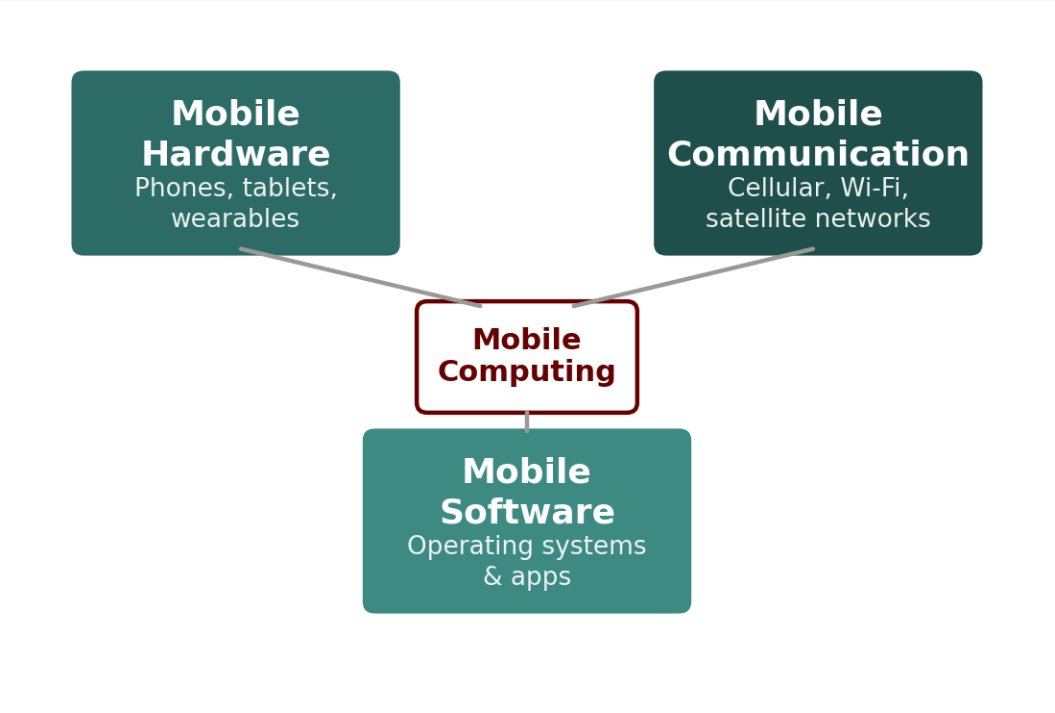


Figure 1: The three core components that together enable mobile computing

- **Mobile hardware:** Smartphones, tablets, wearables — devices with processors, sensors, and radios
- **Mobile software:** Operating systems and applications designed for constrained, portable hardware
- **Mobile communication:** The wireless infrastructure (cellular networks, Wi-Fi, satellite) that carries data between devices and servers

The Generations of Mobile Networks

- **1G (1980s):** Analog voice-only cellular networks; no data capability; poor security and call quality
- **2G (1990s):** Digital voice and basic text messaging (SMS); introduced GSM standard and circuit-switched data
- **3G (2000s):** Packet-switched data enabling mobile internet, video calls, and early smartphones (up to ~2 Mbps)
- **4G / LTE (2010s):** All-IP network design; high-speed mobile broadband (up to ~100 Mbps+) enabling app ecosystems, streaming, and cloud services
- **5G (2020s–present):** Ultra-low latency, massive device density, and multi-Gbps speeds enabling IoT at scale, AR/VR, and real-time applications

Comparing Network Generations

Table 1: Key technical characteristics across mobile network generations

Generation	Era	Peak Speed	Key Capability
1G	1980s	2 kbps	Analog voice calls
2G	1990s	64 kbps	Digital voice + SMS
3G	2000s	~2 Mbps	Mobile internet, video calls
4G / LTE	2010s	~100+ Mbps	Mobile broadband, app ecosystems
5G	2020s	Multi-Gbps	IoT scale, ultra-low latency

Why Network Generation Matters for App Developers

- **Bandwidth assumptions:** Apps must work gracefully across variable network speeds — from poor 3G coverage to 5G
- **Latency-sensitive features:** Real-time apps (gaming, video calls, AR) depend on the low-latency characteristics 5G provides
- **Offline-first design:** Because network quality is never guaranteed, well-designed apps cache data and degrade gracefully when connectivity drops
- **IoT enablement:** 5G's device density support is what makes large-scale IoT integration — a topic in Week 15 — practical

Cellular Network Architecture

Core Components

- **Cell:** A geographic area served by one base station
- **Base Station (BTS/eNodeB):** Transmits/receives radio signals to/from devices in its cell
- **Mobile Switching Center (MSC):** Routes calls and data between cells and the wider network

Why Cells Exist

- **Frequency reuse:** Dividing coverage into small cells lets the same frequencies be reused in non-adjacent cells
- **Capacity scaling:** Smaller cells support more simultaneous users in dense areas
- **Coverage efficiency:** Lower transmission power per cell reduces interference and battery drain

Mobility Management

- **Definition:** The set of network functions that track a mobile device's location and maintain connectivity as the device moves
- **Location management:** The network continuously registers which cell (or area) a device is currently in, so incoming calls/data can be routed correctly
- **Handoff management:** The network transfers an active connection from one cell to another as a device moves, ideally without the user noticing
- **Why it's hard:** Mobility management must operate in real time, at scale, while conserving device battery and network signaling capacity

Handoff (Handover): Keeping Connections Alive

- **Definition**
  - The process of transferring an active call or data session from one base station to another as a device moves between cells
- **Hard handoff**
  - Connection to the old cell is dropped before the new one is established (used in older/2G-style systems)
- **Soft handoff**
  - Device briefly maintains connections to both cells during transition, reducing the risk of dropped calls (used in 3G/4G/5G)
- **Why it matters for apps**
  - A poorly designed app can interpret a brief handoff delay as a network failure and behave incorrectly — robust apps tolerate short interruptions

Roaming

- **Definition:** The ability of a mobile device to remain connected and usable when it moves outside its home network's coverage area, by using another provider's network
- **National roaming:** Using a different network within the same country, typically due to coverage gaps
- **International roaming:** Using a foreign carrier's network when traveling abroad, governed by agreements between operators
- **Practical implication:** Roaming often introduces higher latency, different bandwidth characteristics, and sometimes additional cost — all relevant when designing data-conscious mobile apps

Mobile vs. Wireless vs. Ubiquitous Computing

Table 2: These terms are related but not interchangeable

Term	Core Idea	Example
Mobile computing	Computing that moves with the user	Using a smartphone app on a train
Wireless computing	Computing connected without physical cables	A desktop PC on Wi-Fi (not portable)
Ubiquitous computing	Computing embedded invisibly into everyday environments	Smart home sensors, wearables, ambient displays

Clarifying the Distinctions

- **Mobile computing requires portability:** the device itself must be able to move with the user; this is the defining trait
- **Wireless computing only requires absence of cables:** a stationary device on Wi-Fi is wireless but not mobile
- **Ubiquitous computing ("pervasive computing") goes further:** computing power is distributed invisibly throughout the environment, not concentrated in one personal device
- **Overlap:** Most modern mobile computing is also wireless, and increasingly intersects with ubiquitous computing through IoT — the convergence point we revisit in Week 15

The Mobile Application Ecosystem

- **Devices**
  - Smartphones, tablets, wearables — the hardware running the OS and apps
- **Networks**
  - Cellular and Wi-Fi infrastructure that connects devices to back-end services
- **Platforms**
  - Operating systems (Android, iOS) and their SDKs that define how apps are built and run
- **App stores**
  - Distribution channels (Google Play, Apple App Store) that handle discovery, installation, and monetization
- **Back-end services**
  - Cloud APIs, databases, and authentication services that most non-trivial apps depend on

How the Ecosystem Pieces Connect

Developer's View

- **Build for a platform:** Choose Android, iOS, or cross-platform tools
- **Use device capabilities:** Sensors, camera, GPS, storage
- **Connect to services:** REST APIs, cloud databases (Weeks 7–10)

User's View

- **Discover:** Find the app through an app store or web link
- **Install:** Download and grant permissions
- **Use:** Interact with the app over a network connection, often intermittently

Worked Example: A Ride-Hailing App

- **Mobile computing in action:** The rider's and driver's phones move continuously while staying connected
- **Network generation dependency:** Real-time location updates and matching require low-latency 4G/5G connectivity; performance degrades visibly on 2G/3G
- **Handoff in practice:** A ride across a city involves dozens of handoffs between cells — the app must tolerate brief signal gaps without crashing or losing trip state
- **Ecosystem touchpoints:** The app depends on GPS (device), cellular data (network), the Android/iOS platform (OS), and cloud APIs (back end) simultaneously

Worked Example: Cross-Border Travel

- **Scenario:** A user crosses a national border with their phone powered on and connected
- **Roaming triggered:** The device's home network has no coverage, so it negotiates access to a partner network in the new country (international roaming)
- **Mobility management at play:** Location management must update the network on the new country's infrastructure; handoff occurs continuously as the user moves
- **Design implication:** A well-built travel or messaging app should detect roaming conditions and adapt — for example, warning the user about data costs or switching to a lower-bandwidth mode

Cellular Handoff Illustrated

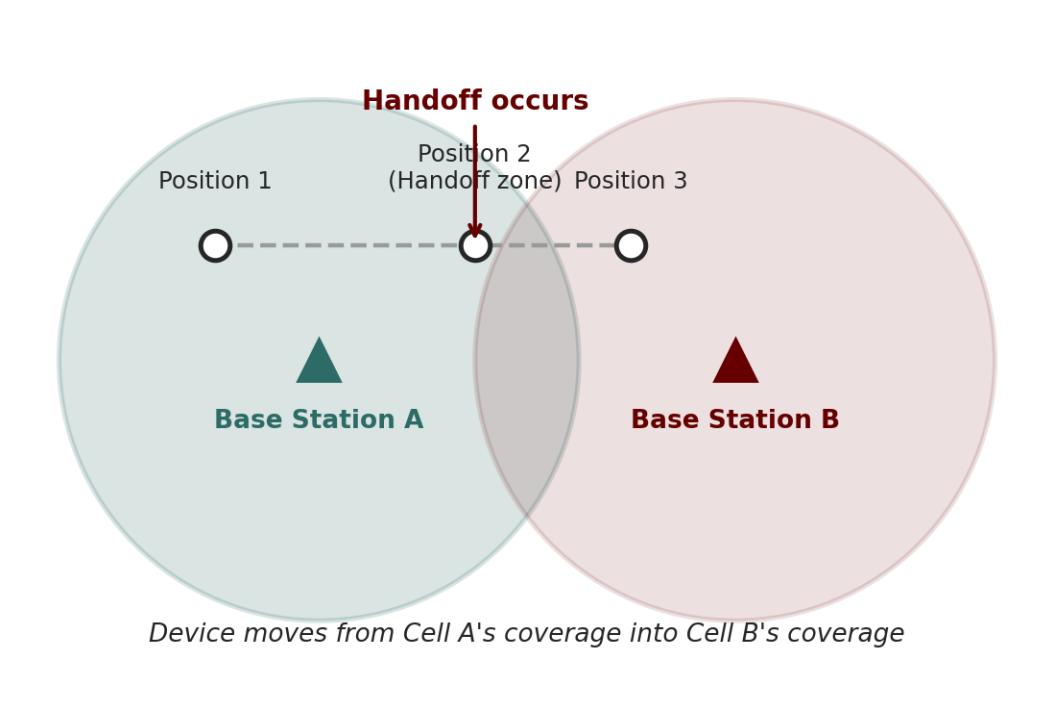


Figure 2: As a device moves from Cell A to Cell B, the network transfers the active connection between base stations

- **Step 1:** Device is connected to Base Station A; signal strength is strong
- **Step 2:** As the device moves, signal from Base Station B becomes stronger than A
- **Step 3:** The network initiates handoff, establishing a connection to B
- **Step 4:** Connection to A is released once B is confirmed active (soft handoff)

Current Trends in Mobile Application Development

- **Cross-platform frameworks:** Flutter and React Native continue to reduce the cost of supporting both Android and iOS from one codebase (explored in Week 3 and Week 12)
- **5G-enabled experiences:** Augmented reality, cloud gaming, and real-time collaboration apps are increasingly viable as 5G rolls out globally
- **On-device AI:** Machine learning models increasingly run directly on mobile hardware rather than purely in the cloud, improving privacy and responsiveness
- **Privacy-first design:** App stores and regulators (e.g., GDPR) are tightening requirements on data collection, consent, and permission transparency
- **IoT convergence:** Mobile apps increasingly serve as the primary control interface for connected devices — the subject of Week 15

Why This Matters for Your Career

- **High demand:** Mobile developers remain among the most sought-after software engineering roles globally
- **Broad applicability:** Skills learned in this module — UI/UX, APIs, databases, security — transfer directly to web and backend development as well
- **Entrepreneurial pathway:** Low barrier to publishing (app stores) makes mobile development a common route into independent software products
- **This module's design:** Each week builds toward Assignment 2 — a complete, deployable mobile application — mirroring how real development teams work

Assessment Plan Overview

Table 3: How your performance in this module will be evaluated

Component	Type	Weight
Assignment 1	Individual	10%
Assignment 2	Group Project	30%
Final Examination	Individual	60%

Module Structure and Practical Expectations

- **Format:** 2 hours of lecture plus 3 hours of practical lab work each week, for 15 weeks
- **Lectures:** Build conceptual understanding — architecture, design principles, security models
- **Practical labs:** Apply that week's concepts hands-on, building toward the skills needed for Assignment 2
- **Assignment 1 (10%):** An individual exercise on a single app component (UI screen, API integration, or sensor feature)
- **Assignment 2 (30%):** A group project building a complete app integrating an API, a database, and a security mechanism
- **Final Examination (60%):** Written assessment covering theoretical understanding and applied problem-solving across the whole module

CONCLUSION

Key Takeaways

- Mobile computing is defined by portability, connectivity, interactivity, individuality, and context-awareness
- Mobile networks evolved from analog voice-only (1G) to multi-Gbps, IoT-scale infrastructure (5G)
- Cellular architecture relies on cell-based frequency reuse, with handoff and roaming maintaining connectivity as devices move
- Mobile, wireless, and ubiquitous computing are related but distinct concepts — portability is mobile computing's defining trait
- The mobile app ecosystem spans devices, networks, platforms, app stores, and back-end services working together

Summary

- This week established the foundational vocabulary and concepts for the entire module: what mobile computing is, how mobile networks have evolved, and how cellular networks keep moving devices connected.
- We distinguished mobile computing from related but different concepts — wireless and ubiquitous computing — and mapped out the ecosystem of devices, networks, platforms, and services that any mobile app depends on.
- These concepts directly underpin next week's topic: the architecture of mobile operating systems themselves.

Discussion Questions

- Think of a mobile app you use daily. Which of the five core characteristics of mobile computing (portability, connectivity, interactivity, individuality, context-awareness) does it rely on most heavily, and why?
- Why might an app that works perfectly on 5G fail or frustrate users on a 3G connection? What design choices could mitigate this?
- Is a smart speaker (e.g., a voice assistant device) an example of mobile computing, wireless computing, or ubiquitous computing? Justify your answer.
- Which part of the mobile app ecosystem (devices, networks, platforms, app stores, back-end services) do you think presents the biggest challenge for a new developer, and why?

Thank You !

Questions?

Email: Sammani@kln.ac.lk | Department of Industrial Management